Visual Search Engine

Submitted by:

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https://github.com/haroon-ai1/cbir-visual-search-engine

Submitted to:

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Feasibility study

Sr.	Requirements	Resources
1	Image Loading and Preprocessing	OpenCV (cv2.imread, cv2.resize, cv2.cvtColor)
2	Color Histogram Feature Extraction	OpenCV (cv2.calcHist, cv2.normalize), NumPy
3	Gabor Filter Texture Feature Extraction	scikit-image (gabor()), OpenCV (cv2.getGaborKernel)
4	Edge-Based Feature Extraction	OpenCV (cv2.Canny, cv2.Sobel), NumPy
5	Combined Feature Vector Generation	NumPy (np.concatenate), Python
6	Feature Database / Index Building	NumPy (np.save, np.load), Python pickle, os.listdir
7	Query Image Input	OpenCV, Python argparse / input(), os.path
8	Similarity Computation	scikit-learn (cosine_similarity), SciPy (cdist), NumPy
9	Top-N Result Retrieval and Display	Matplotlib (plt.subplot, plt.imshow), NumPy (argsort)
10	Scene Enhancement Preprocessing	OpenCV (cv2.createCLAHE, cv2.equalizeHist)

Project Collaboration Summary

Muhammad Haroon (haroon-ai1)

- Muhammad Haroon took the lead on the project's foundation and final deliverables.
- On the technical side, he initialized the GitHub repository, set up the base files.
- developed the Feature Fusion module that successfully combines 70% color and 30% edge features into a unified vector.
- For the project documentation, he authored the main Submission Document, created the GitHub README file, and designed the final PowerPoint (PPT) presentation.

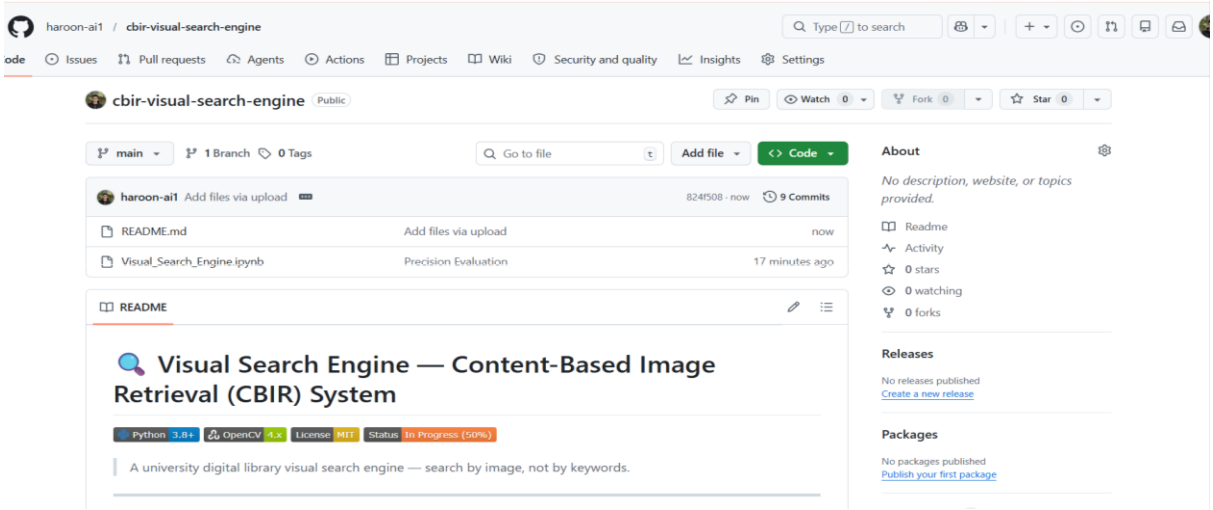
Muneeb Ahmad (MuneebAhmadOfficial)

- Muneeb Ahmad focused heavily on the core feature extraction algorithms.
- successfully implemented the Color Histogram Feature Extraction module
- Developed the Edge Feature Extraction pipeline using Canny edge detection and Sobel gradients.
- Muneeb was instrumental in resolving and fixing integration issues within the feature fusion logic, while also contributing to the collaborative review of the final submission materials.

Mubashir Ahmad (Mibiol)

- Mubashir Ahmad engineered the retrieval and evaluation components of the visual search engine.
- Developed the Query & Retrieve logic, implementing both Cosine Similarity and Euclidean distance metrics to fetch the Top-5 nearest matches.
- Built the Display Results visualization to showcase the query alongside the outputs and programmed the Precision@5 Evaluation metrics to measure the system's retrieval accuracy.
- Assisted in the final review of the project submission.

GitHub Repo: https://github.com/haroon-ai1/cbir-visual-search-engine



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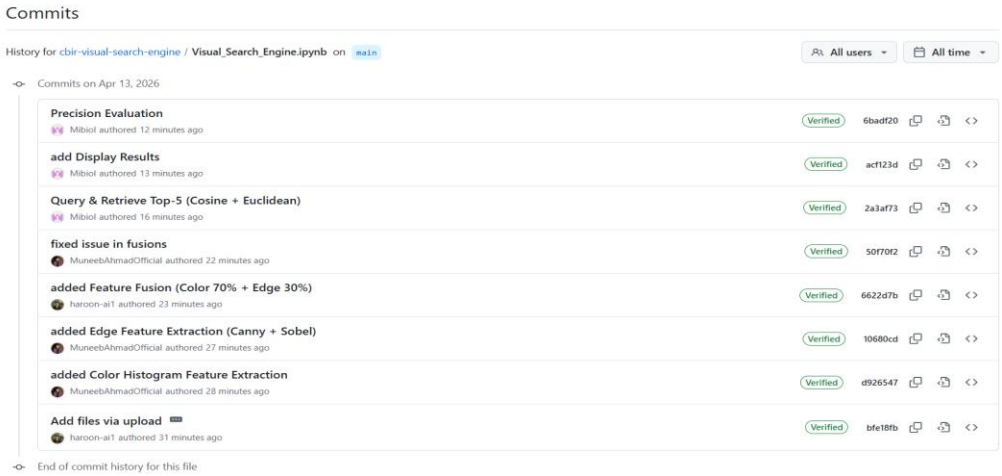
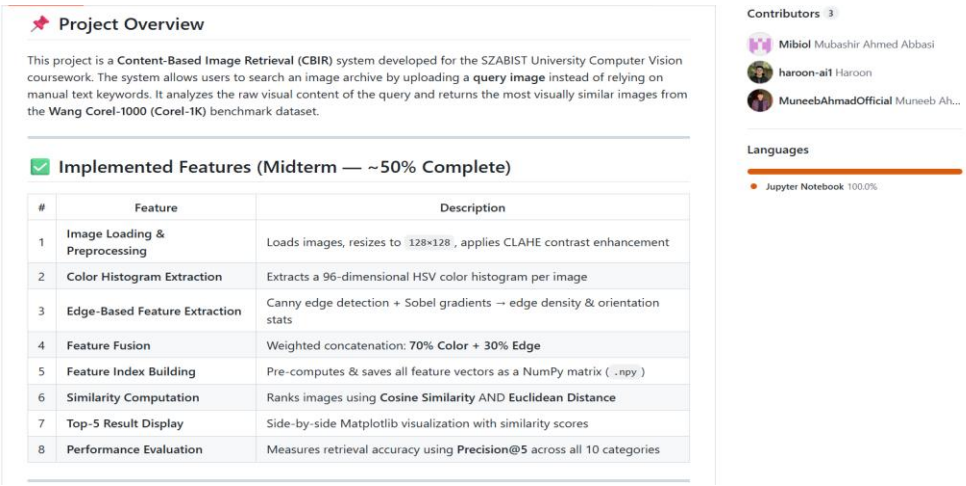


Overview



Project Overview

This project is a **Content-Based Image Retrieval (CBIR)** system developed for the SZABIST University Computer Vision coursework. The system allows users to search an image archive by uploading a **query image** instead of relying on manual text keywords. It analyzes the raw visual content of the query and returns the most visually similar images from the **Wang Corel-1000 (Corel-1K)** benchmark dataset.

Implemented Features (Midterm — ~50% Complete)

#	Feature	Description
1	Image Loading & Preprocessing	Loads images, resizes to `128×128`, applies CLAHE contrast enhancement
2	Color Histogram Extraction	Extracts a 96-dimensional HSV color histogram per image
3	Edge-Based Feature Extraction	Canny edge detection + Sobel gradients → edge density & orientation stats
4	Feature Fusion	Weighted concatenation: **70% Color + 30% Edge**
5	Feature Index Building	Pre-computes & saves all feature vectors as a NumPy matrix (`.npy`)
6	Similarity Computation	Ranks images using **Cosine Similarity** AND **Euclidean Distance**
7	Top-5 Result Display	Side-by-side Matplotlib visualization with similarity scores
8	Performance Evaluation	Measures retrieval accuracy using **Precision@5** across all 10 categories

Contributors 3
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Languages
Jupyter Notebook 100.0%

Tech Stack & contributors

🛠️ Tech Stack

Library	Purpose
`opencv-python`	Image loading, resizing, CLAHE, Canny, Sobel
`numpy`	Feature vectors, matrix operations
`scikit-learn`	Cosine similarity computation
`scipy`	Euclidean distance computation
`matplotlib`	Result visualization

👥 Collaborators

Name	GitHub	Contributions
Muhammad Haroon	@haroon-ai1	Preprocessing, Feature Fusion, Documentation
Muneeb Ahmad	@MuneebAhmadOfficial	Color Histograms, Edge Extraction, Fusion Fixes
Mubashir Ahmad	@Mibiol	Query/Retrieval Logic, Visualizations, Precision Evaluation